

CHITWAN GOEL

Junior Undergraduate | Department of CSE | Indian Institute of Technology Kanpur

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EDUCATION

Indian Institute of Technology Kanpur	2021 - Present
Bachelor of Technology	CGPA: 9.73/10
M.G. Public School, Muzaffarnagar	2021
XII, CBSE	Percentage: 99.0%
M.G. Public School, Muzaffarnagar	2019
X, CBSE	Percentage: 98.8%

KEY PROJECTS

CSE BUBBLE (CS220 Course Project)	Mar '23 - Apr '23
Mentor: Professor Urbi Chatterjee (CSE, IIT Kanpur)	
- Implemented CSE-BUBBLE processor with separate 32-bit sized Instruction and Data Memory sections - Developed op-code formats and modules for instruction handling, including ALU and branching operations - Generated machine code for Bubble Sort from MIPS code, and stored output in data memory through simulation	

Space Travel Booking (CS253 Course Project)	Mar '23
- Built a Space Travel Booking system in C++ that manages the efficient booking of tickets from one planet to other - Created Classes and Objects to implement Object-Oriented Programming for various software functionalities - Added user friendly features like deleting and updating tickets, booking return tickets along with dynamic fare calculation	

What's Next (CS253 Course Project)	Jan '23 - Apr '23
Mentor: Professor Indranil Saha (CSE, IIT Kanpur)	
- Developed an event management MERN app for IITK, featuring registration, like/dislike functionality & notification - Created an intuitive interface with interactive elements like event details pages, and personalized user dashboards - Integrated a payment portal enabling organizers to make direct seamless payments to Lecture Hall Complex	

Jal Jeevan Mission (Water-Life Mission)	Nov '22 - Apr '23
Mentor: Professor Priyanka Bagade (CSE, IIT Kanpur)	
- Contributed to a government-funded project addressing water pipe leakage detection using sensors and hydrophones - Extensively analyzed state-of-the-art techniques and integrated hardware and software approaches in the project - Simulated a water pipeline to collect data and employed KNN-SVM ensemble methods, for accurate predictions	

COURSEWORK

* : Ongoing | A* : Exceptional Performance

Computer Organisation (A*)	Data Structures & Algorithms (A*)
Linear Algebra and ODE	Discrete Mathematics
Fundamentals of Computing	Logic for CS
Probability for CS	Operating Systems*
Software DevOps	Introduction to ML*
Multivariate Calculus	Image Processing*

EXTRA CURRICULARS

Takneek'23	Mar '23 - Apr '23
Led a team of 10 students in the annual Inter-Hall Tech competitions, providing guidance and mentorship on programming problem statements	
Academic Mentor, SatheJee, IIT Kanpur	Feb '22 - Apr '23
Guided and supported JEE aspirants with study materials and helping with their doubts and exam strategies	

WORK EXPERIENCE

AI Intern, AI Northstar Tech	May '23 - Jul '23
- Conducted extensive research on state-of-the-art AI techniques, and used it to solve real-world problems - Curated a terabyte-scale image dataset through web scraping using Selenium and BeautifulSoup, for multiple semantic search engines - Developed a prototype web app for searching and browsing the dataset using images and text as input - Integrated embeddings from large multi-modal models like CLIP, and Transformer models like ViT, into the prototype, using Hugging-face hub and transformers library	
Full Stack Developer Intern, Vizuala	Mar '23 - Jul '23
- Developed a full-stack MERN course portal, featuring integrated payment gateway and user authentication - Implemented a search system using TensorFlow.js, enabling advanced search based on semantic understanding - Automated an uploading process to Firebase, drastically reducing the time required from hours to seconds - Created an innovative resume generation portal utilizing AI and human expertise for polished resumes	

ACHIEVEMENTS & AWARDS

- Secured quant internship position at Quadeye Securities for 2024 - Recipient of the Quadeye Excellence Award 2023, recognized for being among the top 10 students on campus - Conferred with the Academic Excellence Award twice for outstanding academic performance by IIT Kanpur - Got AIR 175 in JEE Advanced 2021 & AIR 155 in JEE Main 2021 - Secured an AIR 837 in the prestigious KVPY fellowship in 2020-21, organized by IISc Bangalore - Among All India Top 1% in Indian Olympiad Qualifier in Physics 2020-21, conducted by HBCSE - Among UP State Top 1% in Indian Olympiad Qualifier in Chemistry 2020-21, conducted by HBCSE - Among Delhi Region Top 60 in Indian Olympiad Qualifier in Mathematics 2020-21, conducted by MTA	
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SKILLS

* : Elementary Exposure

Programming:	C, C++, Python, Bash script, Verilog HDL, MIPS assembly language, Dart*
Frameworks:	Numpy, Pandas, Matplotlib, Keras, Tensorflow, Sklearn, Selenium, Gradui, Streamlit, Pinecone, Weaviate, Qdrant, Flutter*
Web:	HTML, CSS, JavaScript, ReactJS, NodeJS, Express, TypeScript, MongoDB, NextJS, Firebase*
Misc:	Git, LATEX, AutoCAD, Tkinter, Google Colab, Jupyter Notebook

POSITIONS OF RESPONSIBILITY

Web Head, Techkriti, IIT Kanpur	Jun '23 - Present
- Leading a team of senior executives, driving website development with cutting-edge design and user experience - Collaborate with content and marketing teams to align website development with brand strategy and messaging	
Web Incharge, HEC, Hall3, IIT Kanpur	May '23 - Present
Oversee and maintain mess automation portal and hall website, ensuring optimal functionality and user experience	
Company Coordinator, SPO, IIT Kanpur	Jun '22 - Jan '23
Assisted in the internship and placement drive of over 2000 students, ensuring a effective and hassle free process	